Question -3

Code:

```
class question1 {

   public static void main (String[] args) {

      System.out.println("Hello World");
   }
}
```

```
PS D:\CDAC\PGCP-AI\assignment\Assignment -2> javac question1.java
PS D:\CDAC\PGCP-AI\assignment\Assignment -2> java question1
Hello World
```

```
PS D:\CDAC\PGCP-AI\assignment\Assignment -2> javap -c question1
Compiled from "question1.java"
class question1 {
  question1();
    Code:
       0: aload_0
       1: invokespecial #1                  // Method java/lang/Object."<init>":()V
       4: return

  public static void main(java.lang.String[]);
    Code:
       0: getstatic     #2                  // Field java/lang/System.out:Ljava/io/PrintStream;
       3: ldc           #3                  // String Hello World
       5: invokevirtual #4                  // Method java/io/PrintStream.println:(Ljava/lang/String;)V
       8: return
}
PS D:\CDAC\PGCP-AI\assignment\Assignment -2>
```

Question -6

Code:

```
class question2 {
  public static void main (String[] args){
    boolean value = true;
    String output = Boolean.toString(value);

    System.out.println(output);
  }
}
```

```
PS D:\CDAC\PGCP-AI\assignment\Assignment -2> javac question2.java
PS D:\CDAC\PGCP-AI\assignment\Assignment -2> java question2
true
```

```
PS D:\CDAC\PGCP-AI\assignment\Assignment -2> javap -c question2
Compiled from "question2.java"
class question2 {
  question2();
    Code:
       0: aload_0
       1: invokespecial #1                  // Method java/lang/Object."<init>":()V
       4: return

  public static void main(java.lang.String[]);
    Code:
       0: iconst_1
       1: istore_1
       2: iload_1
       3: invokestatic  #2                  // Method java/lang/Boolean.toString:(Z)Ljava/lang/String;
       6: astore_2
       7: getstatic     #3                  // Field java/lang/System.out:Ljava/io/PrintStream;
      10: aload_2
      11: invokevirtual #4                  // Method java/io/PrintStream.println:(Ljava/lang/String;)V
      14: return
}
```

Code:

```
class question3 {
   public static void main(String[]args){
      String name = "Priyansu";
      Boolean output = Boolean.parseBoolean(name);

      System.out.println(output);
   }
}
```

```
PS D:\CDAC\PGCP-AI\assignment\Assignment -2> javac question3.java
PS D:\CDAC\PGCP-AI\assignment\Assignment -2> java question3
false
```

```
PS D:\CDAC\PGCP-AI\assignment\Assignment -2> javap -c question3
Compiled from "question3.java"
class question3 {
  question3();
    Code:
       0: aload_0
       1: invokespecial #1                  // Method java/lang/Object."<init>":()V
       4: return

  public static void main(java.lang.String[]);
    Code:
       0: ldc           #2                  // String Priyansu
       2: astore_1
       3: aload_1
       4: invokestatic  #3                  // Method java/lang/Boolean.parseBoolean:(Ljava/lang/String;)Z
       7: invokestatic  #4                  // Method java/lang/Boolean.valueOf:(Z)Ljava/lang/Boolean;
      10: astore_2
      11: getstatic     #5                  // Field java/lang/System.out:Ljava/io/PrintStream;
      14: aload_2
      15: invokevirtual #6                  // Method java/io/PrintStream.println:(Ljava/lang/Object;)V
      18: return
}
```

Question 7:

Code:

```
class question4 {
 public static void main(String[ ]args){

int bitsvalue = Byte.SIZE;

System.out.println(bitsvalue);

 }
}
```

```
PS D:\CDAC\PGCP-AI\assignment\Assignment -2> javac question4.java
PS D:\CDAC\PGCP-AI\assignment\Assignment -2> java question4
8
```

```
PS D:\CDAC\PGCP-AI\assignment\Assignment -2> javap -c question4
Compiled from "question4.java"
class question4 {
  question4();
    Code:
       0: aload_0
       1: invokespecial #1                  // Method java/lang/Object."<init>":()V
       4: return

  public static void main(java.lang.String[]);
    Code:
       0: bipush        8
       2: istore_1
       3: getstatic     #3                  // Field java/lang/System.out:Ljava/io/PrintStream;
       6: iload_1
       7: invokevirtual #4                  // Method java/io/PrintStream.println:(I)V
      10: return
}
```

Code:

```
public class question5 {
   public static void main (String [] args)
   {
      int bytevalue = Byte.BYTES;

      System.out.println(bytevalue);
   }
}
```

```
PS D:\CDAC\PGCP-AI\assignment\Assignment -2> javac question5.java
PS D:\CDAC\PGCP-AI\assignment\Assignment -2> java question5
1
```

```
PS D:\CDAC\PGCP-AI\assignment\Assignment -2> javap -c question5
Compiled from "question5.java"
public class question5 {
  public question5();
    Code:
       0: aload_0
       1: invokespecial #1                  // Method java/lang/Object."<init>":()V
       4: return

  public static void main(java.lang.String[]);
    Code:
       0: iconst_1
       1: istore_1
       2: getstatic     #3                  // Field java/lang/System.out:Ljava/io/PrintStream;
       5: iload_1
       6: invokevirtual #4                  // Method java/io/PrintStream.println:(I)V
       9: return
}
```

Code:

```
class question6 {
   public static void main(String[ ] args)
   {
int min = Byte.MIN_VALUE;

System.out.println(min);

   }

}
```

```
PS D:\CDAC\PGCP-AI\assignment\Assignment -2> javac question6.java
PS D:\CDAC\PGCP-AI\assignment\Assignment -2> java question6
-128
```

```
PS D:\CDAC\PGCP-AI\assignment\Assignment -2> javap -c question6
Compiled from "question6.java"
class question6 {
  question6();
    Code:
       0: aload_0
       1: invokespecial #1                  // Method java/lang/Object."<init>":()V
       4: return

  public static void main(java.lang.String[]);
    Code:
       0: bipush        -128
       2: istore_1
       3: getstatic     #3                  // Field java/lang/System.out:Ljava/io/PrintStream;
       6: iload_1
       7: invokevirtual #4                  // Method java/io/PrintStream.println:(I)V
      10: return
}
```

Code:

```
class question7 {
   public static void main(String[] args){
      int max = Byte.MAX_VALUE;

      System.out.println(max);
   }
}
```

```
PS D:\CDAC\PGCP-AI\assignment\Assignment -2> javac question7.java
PS D:\CDAC\PGCP-AI\assignment\Assignment -2> java question7
127
```

```
PS D:\CDAC\PGCP-AI\assignment\Assignment -2> javap -c question7
Compiled from "question7.java"
class question7 {
  question7();
    Code:
       0: aload_0
       1: invokespecial #1                  // Method java/lang/Object."<init>":()V
       4: return

  public static void main(java.lang.String[]);
    Code:
       0: bipush        127
       2: istore_1
       3: getstatic     #3                  // Field java/lang/System.out:Ljava/io/PrintStream;
       6: iload_1
       7: invokevirtual #4                  // Method java/io/PrintStream.println:(I)V
      10: return
}
```

Question10:

Code:

```
class question8 {

   public static void main (String[] args){
      int bitschar = Character.SIZE;
      int bytechar = Character.BYTES;

      System.out.println(bitschar);
      System.out.println(bytechar);

   }

}
```

```
PS D:\CDAC\PGCP-AI\assignment\Assignment -2> javac question8.java
PS D:\CDAC\PGCP-AI\assignment\Assignment -2> java question8
16
2
```

```
PS D:\CDAC\PGCP-AI\assignment\Assignment -2> javap -c question8
Compiled from "question8.java"
class question8 {
  question8();
    Code:
       0: aload_0
       1: invokespecial #1                  // Method java/lang/Object."<init>":()V
       4: return

  public static void main(java.lang.String[]);
    Code:
       0: bipush        16
       2: istore_1
       3: iconst_2
       4: istore_2
       5: getstatic     #3                  // Field java/lang/System.out:Ljava/io/PrintStream;
       8: iload_1
       9: invokevirtual #4                  // Method java/io/PrintStream.println:(I)V
      12: getstatic     #3                  // Field java/lang/System.out:Ljava/io/PrintStream;
      15: iload_2
      16: invokevirtual #4                  // Method java/io/PrintStream.println:(I)V
      19: return
}
```

Code:

```
class question9 {

  public static void main(String[] args){
    int char_min = Character.MIN_VALUE;
    int char_max = Character.MAX_VALUE;

    System.out.println(char_min);
    System.out.println(char_max);
  }

}
```

```
PS D:\CDAC\PGCP-AI\assignment\Assignment -2> javac question9.java
PS D:\CDAC\PGCP-AI\assignment\Assignment -2> java question9
0
65535
```

```
PS D:\CDAC\PGCP-AI\assignment\Assignment -2> javap -c question9
Compiled from "question9.java"
class question9 {
  question9();
    Code:
       0: aload_0
       1: invokespecial #1                  // Method java/lang/Object."<init>":()V
       4: return

  public static void main(java.lang.String[]);
    Code:
       0: iconst_0
       1: istore_1
       2: ldc           #3                  // int 65535
       4: istore_2
       5: getstatic     #4                  // Field java/lang/System.out:Ljava/io/PrintStream;
       8: iload_1
       9: invokevirtual #5                  // Method java/io/PrintStream.println:(I)V
      12: getstatic     #4                  // Field java/lang/System.out:Ljava/io/PrintStream;
      15: iload_2
      16: invokevirtual #5                  // Method java/io/PrintStream.println:(I)V
      19: return
}
```

Question 12:

Code:

```
class question10 {
public static void main(String[] args){
   int bits = Short.SIZE;
   System.out.println(bits);
   int bytes = Short.BYTES;
   System.out.println(bytes);
   int minimum = Short.MIN_VALUE;
   System.out.println(minimum);
   int maximum = Short.MAX_VALUE;
   System.out.println(maximum);
}
}
```

```
PS D:\CDAC\PGCP-AI\assignment\Assignment -2> javac question10.java
PS D:\CDAC\PGCP-AI\assignment\Assignment -2> java question10
16
2
-32768
32767
```

```
PS D:\CDAC\PGCP-AI\assignment\Assignment -2> javap -c question10
Compiled from "question10.java"
class question10 {
  question10();
    Code:
       0: aload_0
       1: invokespecial #1                  // Method java/lang/Object."<init>":()V
       4: return

  public static void main(java.lang.String[]);
    Code:
       0: bipush        16
       2: istore_1
       3: getstatic     #3                  // Field java/lang/System.out:Ljava/io/PrintStream;
       6: iload_1
       7: invokevirtual #4                  // Method java/io/PrintStream.println:(I)V
      10: iconst_2
      11: istore_2
      12: getstatic     #3                  // Field java/lang/System.out:Ljava/io/PrintStream;
      15: iload_2
      16: invokevirtual #4                  // Method java/io/PrintStream.println:(I)V
      19: sipush        -32768
      22: istore_3
      23: getstatic     #3                  // Field java/lang/System.out:Ljava/io/PrintStream;
      26: iload_3
      27: invokevirtual #4                  // Method java/io/PrintStream.println:(I)V
      30: sipush        32767
      33: istore        4
      35: getstatic     #3                  // Field java/lang/System.out:Ljava/io/PrintStream;
      38: iload         4
      40: invokevirtual #4                  // Method java/io/PrintStream.println:(I)V
      43: return
}
```

Question 13:

Code:

```
class question11 {
   public static void main (String[] args){

      short num = 59;
      String s = Short.toString(num);

      System.out.println(s);

   }
}
```

```
PS D:\CDAC\PGCP-AI\assignment\Assignment -2> javac question11.java
PS D:\CDAC\PGCP-AI\assignment\Assignment -2> java question11
59
```

```
PS D:\CDAC\PGCP-AI\assignment\Assignment -2> javap -c question11
Compiled from "question11.java"
class question11 {
  question11();
    Code:
       0: aload_0
       1: invokespecial #1                  // Method java/lang/Object."<init>":()V
       4: return

  public static void main(java.lang.String[]);
    Code:
       0: bipush        59
       2: istore_1
       3: iload_1
       4: invokestatic  #2                  // Method java/lang/Short.toString:(S)Ljava/lang/String;
       7: astore_2
       8: getstatic     #3                  // Field java/lang/System.out:Ljava/io/PrintStream;
      11: aload_2
      12: invokevirtual #4                  // Method java/io/PrintStream.println:(Ljava/lang/String;)V
      15: return
}
```

Question 15:

Code:

```
class question12 {

   public static void main (String[] args){

int bits = Integer.SIZE;

System.out.println(bits);

int bytes = Integer.BYTES;
System.out.println(bytes);

int min_value = Integer.MIN_VALUE;
System.out.println(min_value);

int max_value = Integer.MAX_VALUE;
System.out.println(max_value);

   }
}
```

```
PS D:\CDAC\PGCP-AI\assignment\Assignment -2> javac question12.java
PS D:\CDAC\PGCP-AI\assignment\Assignment -2> java question12
32
4
-2147483648
2147483647
```

```
PS D:\CDAC\PGCP-AI\assignment\Assignment -2> javap -c question12
Compiled from "question12.java"
class question12 {
  question12();
    Code:
       0: aload_0
       1: invokespecial #1                  // Method java/lang/Object."<init>":()V
       4: return

  public static void main(java.lang.String[]);
    Code:
       0: bipush        32
       2: istore_1
       3: getstatic     #3                  // Field java/lang/System.out:Ljava/io/PrintStream;
       6: iload_1
       7: invokevirtual #4                  // Method java/io/PrintStream.println:(I)V
      10: iconst_4
      11: istore_2
      12: getstatic     #3                  // Field java/lang/System.out:Ljava/io/PrintStream;
      15: iload_2
      16: invokevirtual #4                  // Method java/io/PrintStream.println:(I)V
      19: ldc           #5                  // int -2147483648
      21: istore_3
      22: getstatic     #3                  // Field java/lang/System.out:Ljava/io/PrintStream;
      25: iload_3
      26: invokevirtual #4                  // Method java/io/PrintStream.println:(I)V
      29: ldc           #6                  // int 2147483647
      31: istore        4
      33: getstatic     #3                  // Field java/lang/System.out:Ljava/io/PrintStream;
      36: iload         4
      38: invokevirtual #4                  // Method java/io/PrintStream.println:(I)V
      41: return
}
```

Question 16:

Code:

```
class question13 {

public static void main(String[] args){
int a = 89;
String b = Integer.toString(a);
System.out.println(b);
int c = 34;
String binary = Integer.toBinaryString(c);
String octal = Integer.toOctalString(c);
String hexadecimal = Integer.toHexString(c);

System.out.println(binary);
System.out.println(octal);
System.out.println(hexadecimal);
}
}
```

```
PS D:\CDAC\PGCP-AI\assignment\Assignment -2> javac question13.java
PS D:\CDAC\PGCP-AI\assignment\Assignment -2> java question13
89
100010
42
22
```

```
PS D:\CDAC\PGCP-AI\assignment\Assignment -2> javap -c question13
Compiled from "question13.java"
class question13 {
  question13();
    Code:
       0: aload_0
       1: invokespecial #1                  // Method java/lang/Object."<init>":()V
       4: return

  public static void main(java.lang.String[]);
    Code:
       0: bipush        89
       2: istore_1
       3: iload_1
       4: invokestatic  #2                  // Method java/lang/Integer.toString:(I)Ljava/lang/String;
       7: astore_2
       8: getstatic     #3                  // Field java/lang/System.out:Ljava/io/PrintStream;
      11: aload_2
      12: invokevirtual #4                  // Method java/io/PrintStream.println:(Ljava/lang/String;)V
      15: bipush        34
      17: istore_3
      18: iload_3
      19: invokestatic  #5                  // Method java/lang/Integer.toBinaryString:(I)Ljava/lang/String;
      22: astore        4
      24: iload_3
      25: invokestatic  #6                  // Method java/lang/Integer.toOctalString:(I)Ljava/lang/String;
      28: astore        5
      30: iload_3
      31: invokestatic  #7                  // Method java/lang/Integer.toHexString:(I)Ljava/lang/String;
      34: astore        6
      36: getstatic     #3                  // Field java/lang/System.out:Ljava/io/PrintStream;
      39: aload         4
      41: invokevirtual #4                  // Method java/io/PrintStream.println:(Ljava/lang/String;)V
      44: getstatic     #3                  // Field java/lang/System.out:Ljava/io/PrintStream;
      47: aload         5
      49: invokevirtual #4                  // Method java/io/PrintStream.println:(Ljava/lang/String;)V
      52: getstatic     #3                  // Field java/lang/System.out:Ljava/io/PrintStream;
      55: aload         6
      57: invokevirtual #4                  // Method java/io/PrintStream.println:(Ljava/lang/String;)V
      60: return
}
```

Question 18:

Code:

```
class question14 {

   public static void main (String[] args){

      int a = 45;
      int b = 74;

      int min = Integer.min(a, b);
      int max = Integer.max(a, b);

      int sum = Integer.sum(a, b);

      System.out.println("Minimum Value: " + min);
      System.out.println("Maximum Value: " +  max);
      System.out.println("Sum:" + sum);

   }

}
```

```
PS D:\CDAC\PGCP-AI\assignment\Assignment -2> javac question14.java
PS D:\CDAC\PGCP-AI\assignment\Assignment -2> java question14
Minimum Value: 45
Maximum Value: 74
Sum:119
```

```
PS D:\CDAC\PGCP-AI\assignment\Assignment -2> javap -c question14
Compiled from "question14.java"
class question14 {
  question14();
    Code:
       0: aload_0
       1: invokespecial #1                  // Method java/lang/Object."<init>":()V
       4: return

  public static void main(java.lang.String[]);
    Code:
       0: bipush        45
       2: istore_1
       3: bipush        74
       5: istore_2
       6: iload_1
       7: iload_2
       8: invokestatic  #2                  // Method java/lang/Integer.min:(II)I
      11: istore_3
      12: iload_1
      13: iload_2
      14: invokestatic  #3                  // Method java/lang/Integer.max:(II)I
      17: istore        4
      19: iload_1
      20: iload_2
      21: invokestatic  #4                  // Method java/lang/Integer.sum:(II)I
      24: istore        5
      26: getstatic     #5                  // Field java/lang/System.out:Ljava/io/PrintStream;
      29: new           #6                  // class java/lang/StringBuilder
      32: dup
      33: invokespecial #7                  // Method java/lang/StringBuilder."<init>":()V
      36: ldc           #8                  // String Minimum Value:
      38: invokevirtual #9                  // Method java/lang/StringBuilder.append:(Ljava/lang/String;)Ljava/lang/StringBuilder;
      41: iload_3
      42: invokevirtual #10                 // Method java/lang/StringBuilder.append:(I)Ljava/lang/StringBuilder;
      45: invokevirtual #11                 // Method java/lang/StringBuilder.toString:()Ljava/lang/String;
      48: invokevirtual #12                 // Method java/io/PrintStream.println:(Ljava/lang/String;)V
      51: getstatic     #5                  // Field java/lang/System.out:Ljava/io/PrintStream;
      54: new           #6                  // class java/lang/StringBuilder
      57: dup
      58: invokespecial #7                  // Method java/lang/StringBuilder."<init>":()V
      61: ldc           #13                 // String Maximum Value:
```


```
      61: ldc           #13                 // String Maximum Value:
```


```
      63: invokevirtual #9                  // Method java/lang/StringBuilder.append:(Ljava/lang/String;)Ljava/lang/StringBuilder;
      66: iload         4
      68: invokevirtual #10                 // Method java/lang/StringBuilder.append:(I)Ljava/lang/StringBuilder;
      71: invokevirtual #11                 // Method java/lang/StringBuilder.toString:()Ljava/lang/String;
      74: invokevirtual #12                 // Method java/io/PrintStream.println:(Ljava/lang/String;)V
      77: getstatic     #5                  // Field java/lang/System.out:Ljava/io/PrintStream;
      80: new           #6                  // class java/lang/StringBuilder
      83: dup
      84: invokespecial #7                  // Method java/lang/StringBuilder."<init>":()V
      87: ldc           #14                 // String Sum:
      89: invokevirtual #9                  // Method java/lang/StringBuilder.append:(Ljava/lang/String;)Ljava/lang/StringBuilder;
      92: iload         5
      94: invokevirtual #10                 // Method java/lang/StringBuilder.append:(I)Ljava/lang/StringBuilder;
      97: invokevirtual #11                 // Method java/lang/StringBuilder.toString:()Ljava/lang/String;
     100: invokevirtual #12                 // Method java/io/PrintStream.println:(Ljava/lang/String;)V
     103: return
}
```

Question19:

Code:

```
class question15 {

public static void main (String[] args){

int bit = Long.SIZE;
int bytes = Long.BYTES;

long min = Long.MIN_VALUE;
long max = Long.MAX_VALUE;

System.out.println("Num of bits:" + bit);
System.out.println("Num of bytes:" + bytes);
System.out.println("Minimum Value:" + min);
System.out.println("Maximum value:" + max);

}

}
```

```
PS D:\CDAC\PGCP-AI\assignment\Assignment -2> javac question15.java
PS D:\CDAC\PGCP-AI\assignment\Assignment -2> java question15
Num of bits:64
Num of bytes:8
Minimum Value:-9223372036854775808
Maximum value:9223372036854775807
```

```
PS D:\CDAC\PGCP-AI\assignment\Assignment -2> javap -c question15
Compiled from "question15.java"
class question15 {
  question15();
    Code:
       0: aload_0
       1: invokespecial #1                  // Method java/lang/Object."<init>":()V
       4: return

  public static void main(java.lang.String[]);
    Code:
       0: bipush        64
       2: istore_1
       3: bipush        8
       5: istore_2
       6: ldc2_w        #3                  // long -9223372036854775808l
       9: lstore_3
      10: ldc2_w        #5                  // long 9223372036854775807l
      13: lstore        5
      15: getstatic     #7                  // Field java/lang/System.out:Ljava/io/PrintStream;
      18: new           #8                  // class java/lang/StringBuilder
      21: dup
      22: invokespecial #9                  // Method java/lang/StringBuilder."<init>":()V
      25: ldc           #10                 // String Num of bits:
      27: invokevirtual #11                 // Method java/lang/StringBuilder.append:(Ljava/lang/String;)Ljava/lang/StringBuilder;
      30: iload_1
      31: invokevirtual #12                 // Method java/lang/StringBuilder.append:(I)Ljava/lang/StringBuilder;
      34: invokevirtual #13                 // Method java/lang/StringBuilder.toString:()Ljava/lang/String;
      37: invokevirtual #14                 // Method java/io/PrintStream.println:(Ljava/lang/String;)V
      40: getstatic     #7                  // Field java/lang/System.out:Ljava/io/PrintStream;
      43: new           #8                  // class java/lang/StringBuilder
      46: dup
      47: invokespecial #9                  // Method java/lang/StringBuilder."<init>":()V
      50: ldc           #15                 // String Num of bytes:
      52: invokevirtual #11                 // Method java/lang/StringBuilder.append:(Ljava/lang/String;)Ljava/lang/StringBuilder;
      55: iload_2
      56: invokevirtual #12                 // Method java/lang/StringBuilder.append:(I)Ljava/lang/StringBuilder;
      59: invokevirtual #13                 // Method java/lang/StringBuilder.toString:()Ljava/lang/String;
      62: invokevirtual #14                 // Method java/io/PrintStream.println:(Ljava/lang/String;)V
      65: getstatic     #7                  // Field java/lang/System.out:Ljava/io/PrintStream;
```


```
      65: getstatic     #7                  // Field java/lang/System.out:Ljava/io/PrintStream;
```


```
      68: new           #8                  // class java/lang/StringBuilder
      71: dup
      72: invokespecial #9                  // Method java/lang/StringBuilder."<init>":()V
      75: ldc           #16                 // String Minimum Value:
      77: invokevirtual #11                 // Method java/lang/StringBuilder.append:(Ljava/lang/String;)Ljava/lang/StringBuilder;
      80: lload_3
      81: invokevirtual #17                 // Method java/lang/StringBuilder.append:(J)Ljava/lang/StringBuilder;
      84: invokevirtual #13                 // Method java/lang/StringBuilder.toString:()Ljava/lang/String;
      87: invokevirtual #14                 // Method java/io/PrintStream.println:(Ljava/lang/String;)V
      90: getstatic     #7                  // Field java/lang/System.out:Ljava/io/PrintStream;
      93: new           #8                  // class java/lang/StringBuilder
      96: dup
      97: invokespecial #9                  // Method java/lang/StringBuilder."<init>":()V
     100: ldc           #18                 // String Maximum value:
     102: invokevirtual #11                 // Method java/lang/StringBuilder.append:(Ljava/lang/String;)Ljava/lang/StringBuilder;
     105: lload         5
     107: invokevirtual #17                 // Method java/lang/StringBuilder.append:(J)Ljava/lang/StringBuilder;
     110: invokevirtual #13                 // Method java/lang/StringBuilder.toString:()Ljava/lang/String;
     113: invokevirtual #14                 // Method java/io/PrintStream.println:(Ljava/lang/String;)V
     116: return
}
```

Question 20:

Code:

```
class question16 {

public static void main(String[] args){

   long a = 1500000L;
String X = Long.toString(a);

System.out.println("Long to String:" + X);

      String binary  = Long.toBinaryString(a);
      String octal   = Long.toOctalString(a);
      String hexadecimal     = Long.toHexString(a);

System.out.println("Binary value:" + binary);
System.out.println("Octal Value:" + octal);
System.out.println("Hexadecimal Value:" + hexadecimal);

}


}
```

```
PS D:\CDAC\PGCP-AI\assignment\Assignment -2> java question16
Long to String:1500000
Binary value:101101110001101100000
Octal Value:5561540
Hexadecimal Value:16e360
```

```
PS D:\CDAC\PGCP-AI\assignment\Assignment -2> javap -c question16
Compiled from "question16.java"
class question16 {
  question16();
    Code:
       0: aload_0
       1: invokespecial #1                  // Method java/lang/Object."<init>":()V
       4: return

  public static void main(java.lang.String[]);
    Code:
       0: ldc2_w        #2                  // long 1500000l
       3: lstore_1
       4: lload_1
       5: invokestatic  #4                  // Method java/lang/Long.toString:(J)Ljava/lang/String;
       8: astore_3
       9: getstatic     #5                  // Field java/lang/System.out:Ljava/io/PrintStream;
      12: new           #6                  // class java/lang/StringBuilder
      15: dup
      16: invokespecial #7                  // Method java/lang/StringBuilder."<init>":()V
      19: ldc           #8                  // String Long to String:
      21: invokevirtual #9                  // Method java/lang/StringBuilder.append:(Ljava/lang/String;)Ljava/lang/StringBuilder;
      24: aload_3
      25: invokevirtual #9                  // Method java/lang/StringBuilder.append:(Ljava/lang/String;)Ljava/lang/StringBuilder;
      28: invokevirtual #10                 // Method java/lang/StringBuilder.toString:()Ljava/lang/String;
      31: invokevirtual #11                 // Method java/io/PrintStream.println:(Ljava/lang/String;)V
      34: lload_1
      35: invokestatic  #12                 // Method java/lang/Long.toBinaryString:(J)Ljava/lang/String;
      38: astore        4
      40: lload_1
      41: invokestatic  #13                 // Method java/lang/Long.toOctalString:(J)Ljava/lang/String;
      44: astore        5
      46: lload_1
      47: invokestatic  #14                 // Method java/lang/Long.toHexString:(J)Ljava/lang/String;
      50: astore        6
      52: getstatic     #5                  // Field java/lang/System.out:Ljava/io/PrintStream;
      55: new           #6                  // class java/lang/StringBuilder
      58: dup
      59: invokespecial #7                  // Method java/lang/StringBuilder."<init>":()V
      62: ldc           #15                 // String Binary value:
      64: invokevirtual #9                  // Method java/lang/StringBuilder.append:(Ljava/lang/String;)Ljava/lang/StringBuilder;
      67: aload         4
      69: invokevirtual #9                  // Method java/lang/StringBuilder.append:(Ljava/lang/String;)Ljava/lang/StringBuilder;
      72: invokevirtual #10                 // Method java/lang/StringBuilder.toString:()Ljava/lang/String;
      75: invokevirtual #11                 // Method java/io/PrintStream.println:(Ljava/lang/String;)V
      78: getstatic     #5                  // Field java/lang/System.out:Ljava/io/PrintStream;
      81: new           #6                  // class java/lang/StringBuilder
      84: dup
      85: invokespecial #7                  // Method java/lang/StringBuilder."<init>":()V
      88: ldc           #16                 // String Octal Value:
      90: invokevirtual #9                  // Method java/lang/StringBuilder.append:(Ljava/lang/String;)Ljava/lang/StringBuilder;
      93: aload         5
      95: invokevirtual #9                  // Method java/lang/StringBuilder.append:(Ljava/lang/String;)Ljava/lang/StringBuilder;
      98: invokevirtual #10                 // Method java/lang/StringBuilder.toString:()Ljava/lang/String;
     101: invokevirtual #11                 // Method java/io/PrintStream.println:(Ljava/lang/String;)V
     104: getstatic     #5                  // Field java/lang/System.out:Ljava/io/PrintStream;
     107: new           #6                  // class java/lang/StringBuilder
     110: dup
     111: invokespecial #7                  // Method java/lang/StringBuilder."<init>":()V
     114: ldc           #17                 // String Hexadecimal Value:
     116: invokevirtual #9                  // Method java/lang/StringBuilder.append:(Ljava/lang/String;)Ljava/lang/StringBuilder;
     119: aload         6
     121: invokevirtual #9                  // Method java/lang/StringBuilder.append:(Ljava/lang/String;)Ljava/lang/StringBuilder;
     124: invokevirtual #10                 // Method java/lang/StringBuilder.toString:()Ljava/lang/String;
     127: invokevirtual #11                 // Method java/io/PrintStream.println:(Ljava/lang/String;)V
     130: return
}
```

Question 22:

Code:

```
class question17 {

   public static void main (String[] args){

      long a = 7463493L;
      long b = 249284L;

      long min = Long.min(a, b);
      long max = Long.max(a, b);
      long sum = Long.sum(a, b);

      System.out.println("Minimum: " + min);
      System.out.println("Maximum: " + max);
      System.out.println("Sum: " + sum);
   }

}
```

```
PS D:\CDAC\PGCP-AI\assignment\Assignment -2> javac question17.java
PS D:\CDAC\PGCP-AI\assignment\Assignment -2> java question17
Minimum: 249284
Maximum: 7463493
Sum: 7712777
```

```
PS D:\CDAC\PGCP-AI\assignment\Assignment -2> javap -c question17
Compiled from "question17.java"
class question17 {
  question17();
    Code:
       0: aload_0
       1: invokespecial #1                  // Method java/lang/Object."<init>":()V
       4: return

  public static void main(java.lang.String[]);
    Code:
       0: ldc2_w        #2                  // long 7463493l
       3: lstore_1
       4: ldc2_w        #4                  // long 249284l
       7: lstore_3
       8: lload_1
       9: lload_3
      10: invokestatic  #6                  // Method java/lang/Long.min:(JJ)J
      13: lstore        5
      15: lload_1
      16: lload_3
      17: invokestatic  #7                  // Method java/lang/Long.max:(JJ)J
      20: lstore        7
      22: lload_1
      23: lload_3
      24: invokestatic  #8                  // Method java/lang/Long.sum:(JJ)J
      27: lstore        9
      29: getstatic     #9                  // Field java/lang/System.out:Ljava/io/PrintStream;
      32: new           #10                 // class java/lang/StringBuilder
      35: dup
      36: invokespecial #11                 // Method java/lang/StringBuilder."<init>":()V
      39: ldc           #12                 // String Minimum:
      41: invokevirtual #13                 // Method java/lang/StringBuilder.append:(Ljava/lang/String;)Ljava/lang/StringBuilder;
      44: lload         5
```

```
      44: lload         5
      46: invokevirtual #14                 // Method java/lang/StringBuilder.append:(J)Ljava/lang/StringBuilder;
      49: invokevirtual #15                 // Method java/lang/StringBuilder.toString:()Ljava/lang/String;
      52: invokevirtual #16                 // Method java/io/PrintStream.println:(Ljava/lang/String;)V
      55: getstatic     #9                  // Field java/lang/System.out:Ljava/io/PrintStream;
      58: new           #10                 // class java/lang/StringBuilder
      61: dup
      62: invokespecial #11                 // Method java/lang/StringBuilder."<init>":()V
      65: ldc           #17                 // String Maximum:
      67: invokevirtual #13                 // Method java/lang/StringBuilder.append:(Ljava/lang/String;)Ljava/lang/StringBuilder;
      70: lload         7
      72: invokevirtual #14                 // Method java/lang/StringBuilder.append:(J)Ljava/lang/StringBuilder;
      75: invokevirtual #15                 // Method java/lang/StringBuilder.toString:()Ljava/lang/String;
      78: invokevirtual #16                 // Method java/io/PrintStream.println:(Ljava/lang/String;)V
      81: getstatic     #9                  // Field java/lang/System.out:Ljava/io/PrintStream;
      84: new           #10                 // class java/lang/StringBuilder
      87: dup
      88: invokespecial #11                 // Method java/lang/StringBuilder."<init>":()V
      91: ldc           #18                 // String Sum:
      93: invokevirtual #13                 // Method java/lang/StringBuilder.append:(Ljava/lang/String;)Ljava/lang/StringBuilder;
      96: lload         9
      98: invokevirtual #14                 // Method java/lang/StringBuilder.append:(J)Ljava/lang/StringBuilder;
     101: invokevirtual #15                 // Method java/lang/StringBuilder.toString:()Ljava/lang/String;
     104: invokevirtual #16                 // Method java/io/PrintStream.println:(Ljava/lang/String;)V
     107: return
}
```

Question 23:

Code:

```
class question18 {

   public static void main (String[] args){

      int bit = Float.SIZE;
      int bytes = Float.BYTES;
      float min_value = Float.MIN_VALUE;
      float max_value = Float.MAX_VALUE;

System.out.println("Num of bit: " + bit);
System.out.println("Num of bytes: " + bytes);
System.out.println("Minimum Value: " + min_value);
System.out.println("Maximum Value: " + max_value);

   }

}
```

```
PS D:\CDAC\PGCP-AI\assignment\Assignment -2> javac question18.java
PS D:\CDAC\PGCP-AI\assignment\Assignment -2> java question18
Num of bit: 32
Num of bytes: 4
Minimum Value: 1.4E-45
Maximum Value: 3.4028235E38
```

```
PS D:\CDAC\PGCP-AI\assignment\Assignment -2> javap -c question18
Compiled from "question18.java"
class question18 {
  question18();
    Code:
       0: aload_0
       1: invokespecial #1                  // Method java/lang/Object."<init>":()V
       4: return

  public static void main(java.lang.String[]);
    Code:
       0: bipush        32
       2: istore_1
       3: iconst_4
       4: istore_2
       5: ldc           #3                  // float 1.4E-45f
       7: fstore_3
       8: ldc           #4                  // float 3.4028235E38f
      10: fstore        4
      12: getstatic     #5                  // Field java/lang/System.out:Ljava/io/PrintStream;
      15: new           #6                  // class java/lang/StringBuilder
      18: dup
      19: invokespecial #7                  // Method java/lang/StringBuilder."<init>":()V
      22: ldc           #8                  // String Num of bit:
      24: invokevirtual #9                  // Method java/lang/StringBuilder.append:(Ljava/lang/String;)Ljava/lang/StringBuilder;
      27: iload_1
      28: invokevirtual #10                 // Method java/lang/StringBuilder.append:(I)Ljava/lang/StringBuilder;
      31: invokevirtual #11                 // Method java/lang/StringBuilder.toString:()Ljava/lang/String;
      34: invokevirtual #12                 // Method java/io/PrintStream.println:(Ljava/lang/String;)V
      37: getstatic     #5                  // Field java/lang/System.out:Ljava/io/PrintStream;
      40: new           #6                  // class java/lang/StringBuilder
      43: dup
      44: invokespecial #7                  // Method java/lang/StringBuilder."<init>":()V
```

```
      44: invokespecial #7                  // Method java/lang/StringBuilder."<init>":()V
      47: ldc           #13                 // String Num of bytes:
      49: invokevirtual #9                  // Method java/lang/StringBuilder.append:(Ljava/lang/String;)Ljava/lang/StringBuilder;
      52: iload_2
      53: invokevirtual #10                 // Method java/lang/StringBuilder.append:(I)Ljava/lang/StringBuilder;
      56: invokevirtual #11                 // Method java/lang/StringBuilder.toString:()Ljava/lang/String;
      59: invokevirtual #12                 // Method java/io/PrintStream.println:(Ljava/lang/String;)V
      62: getstatic     #5                  // Field java/lang/System.out:Ljava/io/PrintStream;
      65: new           #6                  // class java/lang/StringBuilder
      68: dup
      69: invokespecial #7                  // Method java/lang/StringBuilder."<init>":()V
      72: ldc           #14                 // String Minimum Value:
      74: invokevirtual #9                  // Method java/lang/StringBuilder.append:(Ljava/lang/String;)Ljava/lang/StringBuilder;
      77: fload_3
      78: invokevirtual #15                 // Method java/lang/StringBuilder.append:(F)Ljava/lang/StringBuilder;
      81: invokevirtual #11                 // Method java/lang/StringBuilder.toString:()Ljava/lang/String;
      84: invokevirtual #12                 // Method java/io/PrintStream.println:(Ljava/lang/String;)V
      87: getstatic     #5                  // Field java/lang/System.out:Ljava/io/PrintStream;
      90: new           #6                  // class java/lang/StringBuilder
      93: dup
      94: invokespecial #7                  // Method java/lang/StringBuilder."<init>":()V
      97: ldc           #16                 // String Maximum Value:
      99: invokevirtual #9                  // Method java/lang/StringBuilder.append:(Ljava/lang/String;)Ljava/lang/StringBuilder;
     102: fload         4
     104: invokevirtual #15                 // Method java/lang/StringBuilder.append:(F)Ljava/lang/StringBuilder;
     107: invokevirtual #11                 // Method java/lang/StringBuilder.toString:()Ljava/lang/String;
     110: invokevirtual #12                 // Method java/io/PrintStream.println:(Ljava/lang/String;)V
     113: return
}
```

Question 24:

Code:

```
class question19 {

   public static void main(String[] args){

      float A = 3.84f;

      String B = Float.toString(A);
      String C = Float.toHexString(A);

      System.out.println("Float to String: " + B);
      System.out.println("Float to Hexadecimal: " + C);

   }

}
```

```
PS D:\CDAC\PGCP-AI\assignment\Assignment -2> javac question19.java
PS D:\CDAC\PGCP-AI\assignment\Assignment -2> java question19
Float to String: 3.84
Float to Hexadecimal: 0x1.eb851ep1
```

```
PS D:\CDAC\PGCP-AI\assignment\Assignment -2> javap -c question19
Compiled from "question19.java"
class question19 {
  question19();
    Code:
       0: aload_0
       1: invokespecial #1                  // Method java/lang/Object."<init>":()V
       4: return

  public static void main(java.lang.String[]);
    Code:
       0: ldc           #2                  // float 3.84f
       2: fstore_1
       3: fload_1
       4: invokestatic  #3                  // Method java/lang/Float.toString:(F)Ljava/lang/String;
       7: astore_2
       8: fload_1
       9: invokestatic  #4                  // Method java/lang/Float.toHexString:(F)Ljava/lang/String;
      12: astore_3
      13: getstatic     #5                  // Field java/lang/System.out:Ljava/io/PrintStream;
      16: new           #6                  // class java/lang/StringBuilder
      19: dup
      20: invokespecial #7                  // Method java/lang/StringBuilder."<init>":()V
      23: ldc           #8                  // String Float to String:
      25: invokevirtual #9                  // Method java/lang/StringBuilder.append:(Ljava/lang/String;)Ljava/lang/StringBuilder;
      28: aload_2
      29: invokevirtual #9                  // Method java/lang/StringBuilder.append:(Ljava/lang/String;)Ljava/lang/StringBuilder;
      32: invokevirtual #10                 // Method java/lang/StringBuilder.toString:()Ljava/lang/String;
      35: invokevirtual #11                 // Method java/io/PrintStream.println:(Ljava/lang/String;)V
      38: getstatic     #5                  // Field java/lang/System.out:Ljava/io/PrintStream;
      41: new           #6                  // class java/lang/StringBuilder
```

```
      41: new           #6                  // class java/lang/StringBuilder
      44: dup
      45: invokespecial #7                  // Method java/lang/StringBuilder."<init>":()V
      48: ldc           #12                 // String Float to Hexadecimal:
      50: invokevirtual #9                  // Method java/lang/StringBuilder.append:(Ljava/lang/String;)Ljava/lang/StringBuilder;
      53: aload_3
      54: invokevirtual #9                  // Method java/lang/StringBuilder.append:(Ljava/lang/String;)Ljava/lang/StringBuilder;
      57: invokevirtual #10                 // Method java/lang/StringBuilder.toString:()Ljava/lang/String;
      60: invokevirtual #11                 // Method java/io/PrintStream.println:(Ljava/lang/String;)V
      63: return
}
```

Question 26:

Code:

```
class question20 {

   public static void main (String[] args){

      float F1 = 3.25f;
      float F2 = 6.37f;

      float Min_Value = Float.min(F1, F2);
      float Max_Value = Float.max(F1, F2);

      float sum = Float.sum(F1, F2);

      System.out.println("Minimum Value:" + Min_Value);
      System.out.println("Maximum Value:" + Max_Value);
      System.out.println("Sum:" + sum);

   }

}
```

```
PS D:\CDAC\PGCP-AI\assignment\Assignment -2> javac question20.java
PS D:\CDAC\PGCP-AI\assignment\Assignment -2> java question20
Minimum Value:3.25
Maximum Value:6.37
Sum:9.62
```

```
PS D:\CDAC\PGCP-AI\assignment\Assignment -2> javap -c question20
Compiled from "question20.java"
class question20 {
  question20();
    Code:
       0: aload_0
       1: invokespecial #1                  // Method java/lang/Object."<init>":()V
       4: return

  public static void main(java.lang.String[]);
    Code:
       0: ldc           #2                  // float 3.25f
       2: fstore_1
       3: ldc           #3                  // float 6.37f
       5: fstore_2
       6: fload_1
       7: fload_2
       8: invokestatic  #4                  // Method java/lang/Float.min:(FF)F
      11: fstore_3
      12: fload_1
      13: fload_2
      14: invokestatic  #5                  // Method java/lang/Float.max:(FF)F
      17: fstore        4
      19: fload_1
      20: fload_2
      21: invokestatic  #6                  // Method java/lang/Float.sum:(FF)F
      24: fstore        5
      26: getstatic     #7                  // Field java/lang/System.out:Ljava/io/PrintStream;
      29: new           #8                  // class java/lang/StringBuilder
      32: dup
      33: invokespecial #9                  // Method java/lang/StringBuilder."<init>":()V
      36: ldc           #10                 // String Minimum Value:
      38: invokevirtual #11                 // Method java/lang/StringBuilder.append:(Ljava/lang/String;)Ljava/lang/StringBuilder;
      41: fload_3
      42: invokevirtual #12                 // Method java/lang/StringBuilder.append:(F)Ljava/lang/StringBuilder;
      45: invokevirtual #13                 // Method java/lang/StringBuilder.toString:()Ljava/lang/String;
```

```
      45: invokevirtual #13                 // Method java/lang/StringBuilder.toString:()Ljava/lang/String;
      48: invokevirtual #14                 // Method java/io/PrintStream.println:(Ljava/lang/String;)V
      51: getstatic     #7                  // Field java/lang/System.out:Ljava/io/PrintStream;
      54: new           #8                  // class java/lang/StringBuilder
      57: dup
      58: invokespecial #9                  // Method java/lang/StringBuilder."<init>":()V
      61: ldc           #15                 // String Maximum Value:
      63: invokevirtual #11                 // Method java/lang/StringBuilder.append:(Ljava/lang/String;)Ljava/lang/StringBuilder;
      66: fload         4
      68: invokevirtual #12                 // Method java/lang/StringBuilder.append:(F)Ljava/lang/StringBuilder;
      71: invokevirtual #13                 // Method java/lang/StringBuilder.toString:()Ljava/lang/String;
      74: invokevirtual #14                 // Method java/io/PrintStream.println:(Ljava/lang/String;)V
      77: getstatic     #7                  // Field java/lang/System.out:Ljava/io/PrintStream;
      80: new           #8                  // class java/lang/StringBuilder
      83: dup
      84: invokespecial #9                  // Method java/lang/StringBuilder."<init>":()V
      87: ldc           #16                 // String Sum:
      89: invokevirtual #11                 // Method java/lang/StringBuilder.append:(Ljava/lang/String;)Ljava/lang/StringBuilder;
      92: fload         5
      94: invokevirtual #12                 // Method java/lang/StringBuilder.append:(F)Ljava/lang/StringBuilder;
      97: invokevirtual #13                 // Method java/lang/StringBuilder.toString:()Ljava/lang/String;
     100: invokevirtual #14                 // Method java/io/PrintStream.println:(Ljava/lang/String;)V
     103: return
}
```

Question 27:

Code:

```
class question21 {

   public static void main (String[] args){

      int bit = Double.SIZE;
      int bytes = Double.BYTES;
      double min_value = Double.MIN_VALUE;
      double max_value = Double.MAX_VALUE;

      System.out.println("Bit Value:" +bit);
      System.out.println("Byte Value:" + bytes);
      System.out.println("Minimum Value:" + min_value);
      System.out.println("Maximum Value:" + max_value);
   }

}
```

```
PS D:\CDAC\PGCP-AI\assignment\Assignment -2> javac question21.java
PS D:\CDAC\PGCP-AI\assignment\Assignment -2> java question21
Bit Value:64
Byte Value:8
Minimum Value:4.9E-324
Maximum Value:1.7976931348623157E308
```

```
PS D:\CDAC\PGCP-AI\assignment\Assignment -2> javap -c question21
Compiled from "question21.java"
class question21 {
  question21();
    Code:
       0: aload_0
       1: invokespecial #1                  // Method java/lang/Object."<init>":()V
       4: return

  public static void main(java.lang.String[]);
    Code:
       0: bipush        64
       2: istore_1
       3: bipush        8
       5: istore_2
       6: ldc2_w        #3                  // double 4.9E-324d
       9: dstore_3
      10: ldc2_w        #5                  // double 1.7976931348623157E308d
      13: dstore        5
      15: getstatic     #7                  // Field java/lang/System.out:Ljava/io/PrintStream;
      18: new           #8                  // class java/lang/StringBuilder
      21: dup
      22: invokespecial #9                  // Method java/lang/StringBuilder."<init>":()V
      25: ldc           #10                 // String Bit Value:
      27: invokevirtual #11                 // Method java/lang/StringBuilder.append:(Ljava/lang/String;)Ljava/lang/StringBuilder;
      30: iload_1
      31: invokevirtual #12                 // Method java/lang/StringBuilder.append:(I)Ljava/lang/StringBuilder;
      34: invokevirtual #13                 // Method java/lang/StringBuilder.toString:()Ljava/lang/String;
      37: invokevirtual #14                 // Method java/io/PrintStream.println:(Ljava/lang/String;)V
      40: getstatic     #7                  // Field java/lang/System.out:Ljava/io/PrintStream;
      43: new           #8                  // class java/lang/StringBuilder
      46: dup
      47: invokespecial #9                  // Method java/lang/StringBuilder."<init>":()V
      50: ldc           #15                 // String Byte Value:
      52: invokevirtual #11                 // Method java/lang/StringBuilder.append:(Ljava/lang/String;)Ljava/lang/StringBuilder;
      55: iload_2
      56: invokevirtual #12                 // Method java/lang/StringBuilder.append:(I)Ljava/lang/StringBuilder;
      59: invokevirtual #13                 // Method java/lang/StringBuilder.toString:()Ljava/lang/String;
      62: invokevirtual #14                 // Method java/io/PrintStream.println:(Ljava/lang/String;)V
      65: getstatic     #7                  // Field java/lang/System.out:Ljava/io/PrintStream;
```

```
      62: invokevirtual #14                 // Method java/io/PrintStream.println:(Ljava/lang/String;)V
      65: getstatic     #7                  // Field java/lang/System.out:Ljava/io/PrintStream;
      68: new           #8                  // class java/lang/StringBuilder
      71: dup
      72: invokespecial #9                  // Method java/lang/StringBuilder."<init>":()V
      75: ldc           #16                 // String Minimum Value:
      77: invokevirtual #11                 // Method java/lang/StringBuilder.append:(Ljava/lang/String;)Ljava/lang/StringBuilder;
      80: dload_3
      81: invokevirtual #17                 // Method java/lang/StringBuilder.append:(D)Ljava/lang/StringBuilder;
      84: invokevirtual #13                 // Method java/lang/StringBuilder.toString:()Ljava/lang/String;
      87: invokevirtual #14                 // Method java/io/PrintStream.println:(Ljava/lang/String;)V
      90: getstatic     #7                  // Field java/lang/System.out:Ljava/io/PrintStream;
      93: new           #8                  // class java/lang/StringBuilder
      96: dup
      97: invokespecial #9                  // Method java/lang/StringBuilder."<init>":()V
     100: ldc           #18                 // String Maximum Value:
     102: invokevirtual #11                 // Method java/lang/StringBuilder.append:(Ljava/lang/String;)Ljava/lang/StringBuilder;
     105: dload         5
     107: invokevirtual #17                 // Method java/lang/StringBuilder.append:(D)Ljava/lang/StringBuilder;
     110: invokevirtual #13                 // Method java/lang/StringBuilder.toString:()Ljava/lang/String;
     113: invokevirtual #14                 // Method java/io/PrintStream.println:(Ljava/lang/String;)V
     116: return
}
```

Question 28:

Code:

```
class question22 {

   public static void main (String[] args){

      double A = 4.8;

      String Num = Double.toString(A);
      long bits = Double.doubleToLongBits(A);

      String binary  = Long.toBinaryString(bits);
      String octal  = Long.toOctalString(bits);
      String hexdecimal  = Long.toHexString(bits);

      System.out.println("Double to string value:" + Num);
      System.out.println("Binary Value:" +binary);
      System.out.println("Octal Value:" + octal);
      System.out.println("Hexadecimal Value:" + hexdecimal);


   }

}
```

```
PS D:\CDAC\PGCP-AI\assignment\Assignment -2> javac question22.java
PS D:\CDAC\PGCP-AI\assignment\Assignment -2> java question22
Double to string value:4.8
Binary Value:1000000000100110011001100110011001100110011001100110011001100110011
Octal Value:400231463146314631463
Hexadecimal Value:4013333333333333
```

```
PS D:\CDAC\PGCP-AI\assignment\Assignment -2> javap -c question22
Compiled from "question22.java"
class question22 {
  question22();
    Code:
       0: aload_0
       1: invokespecial #1                  // Method java/lang/Object."<init>":()V
       4: return

  public static void main(java.lang.String[]);
    Code:
       0: ldc2_w        #2                  // double 4.8d
       3: dstore_1
       4: dload_1
       5: invokestatic  #4                  // Method java/lang/Double.toString:(D)Ljava/lang/String;
       8: astore_3
       9: dload_1
      10: invokestatic  #5                  // Method java/lang/Double.doubleToLongBits:(D)J
      13: lstore        4
      15: lload         4
      17: invokestatic  #6                  // Method java/lang/Long.toBinaryString:(J)Ljava/lang/String;
      20: astore        6
      22: lload         4
      24: invokestatic  #7                  // Method java/lang/Long.toOctalString:(J)Ljava/lang/String;
      27: astore        7
      29: lload         4
      31: invokestatic  #8                  // Method java/lang/Long.toHexString:(J)Ljava/lang/String;
      34: astore        8
      36: getstatic     #9                  // Field java/lang/System.out:Ljava/io/PrintStream;
      39: new           #10                 // class java/lang/StringBuilder
      42: dup
      43: invokespecial #11                 // Method java/lang/StringBuilder."<init>":()V
      46: ldc           #12                 // String Double to string value:
      48: invokevirtual #13                 // Method java/lang/StringBuilder.append:(Ljava/lang/String;)Ljava/lang/StringBuilder;
      51: aload_3
      52: invokevirtual #13                 // Method java/lang/StringBuilder.append:(Ljava/lang/String;)Ljava/lang/StringBuilder;
      55: invokevirtual #14                 // Method java/lang/StringBuilder.toString:()Ljava/lang/String;
      58: invokevirtual #15                 // Method java/io/PrintStream.println:(Ljava/lang/String;)V
      61: getstatic     #9                  // Field java/lang/System.out:Ljava/io/PrintStream;
      64: new           #10                 // class java/lang/StringBuilder
```

```
      64: new           #10                 // class java/lang/StringBuilder
      67: dup
      68: invokespecial #11                 // Method java/lang/StringBuilder."<init>":()V
      71: ldc           #16                 // String Binary Value:
      73: invokevirtual #13                 // Method java/lang/StringBuilder.append:(Ljava/lang/String;)Ljava/lang/StringBuilder;
      76: aload         6
      78: invokevirtual #13                 // Method java/lang/StringBuilder.append:(Ljava/lang/String;)Ljava/lang/StringBuilder;
      81: invokevirtual #14                 // Method java/lang/StringBuilder.toString:()Ljava/lang/String;
      84: invokevirtual #15                 // Method java/io/PrintStream.println:(Ljava/lang/String;)V
      87: getstatic     #9                  // Field java/lang/System.out:Ljava/io/PrintStream;
      90: new           #10                 // class java/lang/StringBuilder
      93: dup
      94: invokespecial #11                 // Method java/lang/StringBuilder."<init>":()V
      97: ldc           #17                 // String Octal Value:
      99: invokevirtual #13                 // Method java/lang/StringBuilder.append:(Ljava/lang/String;)Ljava/lang/StringBuilder;
     102: aload         7
     104: invokevirtual #13                 // Method java/lang/StringBuilder.append:(Ljava/lang/String;)Ljava/lang/StringBuilder;
     107: invokevirtual #14                 // Method java/lang/StringBuilder.toString:()Ljava/lang/String;
     110: invokevirtual #15                 // Method java/io/PrintStream.println:(Ljava/lang/String;)V
     113: getstatic     #9                  // Field java/lang/System.out:Ljava/io/PrintStream;
     116: new           #10                 // class java/lang/StringBuilder
     119: dup
     120: invokespecial #11                 // Method java/lang/StringBuilder."<init>":()V
     123: ldc           #18                 // String Hexadecimal Value:
     125: invokevirtual #13                 // Method java/lang/StringBuilder.append:(Ljava/lang/String;)Ljava/lang/StringBuilder;
     128: aload         8
     130: invokevirtual #13                 // Method java/lang/StringBuilder.append:(Ljava/lang/String;)Ljava/lang/StringBuilder;
     133: invokevirtual #14                 // Method java/lang/StringBuilder.toString:()Ljava/lang/String;
     136: invokevirtual #15                 // Method java/io/PrintStream.println:(Ljava/lang/String;)V
     139: return
}
```

Question 30:

Code:

```
class question23 {

   public static void main(String[] args){

      double D1 = 3.53535;
      double D2 = 6.33434;

      double min_value = Double.min(D1, D2);
      double max_value = Double.max(D1, D2);
      double sum = Double.sum(D1, D2);

      System.out.println("Minimum Value:" + min_value);
      System.out.println("Maximum Value:" + max_value);
      System.out.println("Sum:" + sum);
   }

}
```

```
PS D:\CDAC\PGCP-AI\assignment\Assignment -2> javac question23.java
PS D:\CDAC\PGCP-AI\assignment\Assignment -2> java question23
Minimum Value:3.53535
Maximum Value:6.33434
Sum:9.86969
```

```
PS D:\CDAC\PGCP-AI\assignment\Assignment -2> javap -c question23
Compiled from "question23.java"
class question23 {
  question23();
    Code:
       0: aload_0
       1: invokespecial #1                  // Method java/lang/Object."<init>":()V
       4: return

  public static void main(java.lang.String[]);
    Code:
       0: ldc2_w        #2                  // double 3.53535d
       3: dstore_1
       4: ldc2_w        #4                  // double 6.33434d
       7: dstore_3
       8: dload_1
       9: dload_3
      10: invokestatic  #6                  // Method java/lang/Double.min:(DD)D
      13: dstore        5
      15: dload_1
      16: dload_3
      17: invokestatic  #7                  // Method java/lang/Double.max:(DD)D
      20: dstore        7
      22: dload_1
      23: dload_3
      24: invokestatic  #8                  // Method java/lang/Double.sum:(DD)D
      27: dstore        9
      29: getstatic     #9                  // Field java/lang/System.out:Ljava/io/PrintStream;
      32: new           #10                 // class java/lang/StringBuilder
      35: dup
      36: invokespecial #11                 // Method java/lang/StringBuilder."<init>":()V
      39: ldc           #12                 // String Minimum Value:
      41: invokevirtual #13                 // Method java/lang/StringBuilder.append:(Ljava/lang/String;)Ljava/lang/StringBuilder;
      44: dload         5
      46: invokevirtual #14                 // Method java/lang/StringBuilder.append:(D)Ljava/lang/StringBuilder;
      49: invokevirtual #15                 // Method java/lang/StringBuilder.toString:()Ljava/lang/String;
      52: invokevirtual #16                 // Method java/io/PrintStream.println:(Ljava/lang/String;)V
      55: getstatic     #9                  // Field java/lang/System.out:Ljava/io/PrintStream;
```

```
      55: getstatic     #9                  // Field java/lang/System.out:Ljava/io/PrintStream;
      58: new           #10                 // class java/lang/StringBuilder
      61: dup
      62: invokespecial #11                 // Method java/lang/StringBuilder."<init>":()V
      65: ldc           #17                 // String Maximum Value:
      67: invokevirtual #13                 // Method java/lang/StringBuilder.append:(Ljava/lang/String;)Ljava/lang/StringBuilder;
      70: dload         7
      72: invokevirtual #14                 // Method java/lang/StringBuilder.append:(D)Ljava/lang/StringBuilder;
      75: invokevirtual #15                 // Method java/lang/StringBuilder.toString:()Ljava/lang/String;
      78: invokevirtual #16                 // Method java/io/PrintStream.println:(Ljava/lang/String;)V
      81: getstatic     #9                  // Field java/lang/System.out:Ljava/io/PrintStream;
      84: new           #10                 // class java/lang/StringBuilder
      87: dup
      88: invokespecial #11                 // Method java/lang/StringBuilder."<init>":()V
      91: ldc           #18                 // String Sum:
      93: invokevirtual #13                 // Method java/lang/StringBuilder.append:(Ljava/lang/String;)Ljava/lang/StringBuilder;
      96: dload         9
      98: invokevirtual #14                 // Method java/lang/StringBuilder.append:(D)Ljava/lang/StringBuilder;
     101: invokevirtual #15                 // Method java/lang/StringBuilder.toString:()Ljava/lang/String;
     104: invokevirtual #16                 // Method java/io/PrintStream.println:(Ljava/lang/String;)V
     107: return
}
```

Question 6b:

Code:

```
class question25 {

   public static void main(String[] args){
   boolean B1 = true;
   Boolean B2 = Boolean.valueOf(B1);

   System.out.println(B2);
   }

}
```

```
PS D:\CDAC\PGCP-AI\assignment\Assignment -2> javac question25.java
PS D:\CDAC\PGCP-AI\assignment\Assignment -2> java question25
true
```

```
PS D:\CDAC\PGCP-AI\assignment\Assignment -2> javap -c question25
Compiled from "question25.java"
class question25 {
  question25();
    Code:
       0: aload_0
       1: invokespecial #1                  // Method java/lang/Object."<init>":()V
       4: return

  public static void main(java.lang.String[]);
    Code:
       0: iconst_1
       1: istore_1
       2: new           #2                  // class java/lang/Boolean
       5: dup
       6: iload_1
       7: invokespecial #3                  // Method java/lang/Boolean."<init>":(Z)V
      10: astore_2
      11: getstatic     #4                  // Field java/lang/System.out:Ljava/io/PrintStream;
      14: aload_2
      15: invokevirtual #5                  // Method java/io/PrintStream.println:(Ljava/lang/Object;)V
      18: return
}
```

Question 6d:

Code:

```
class question26 {
   public static void main(String[] args){

      String a = "priyansu";

      Boolean b = Boolean.valueOf(a);

      System.out.print(b);

   }
}
```

```
PS D:\CDAC\PGCP-AI\assignment\Assignment -2> javac question26.java
PS D:\CDAC\PGCP-AI\assignment\Assignment -2> java question26
false
```

```
PS D:\CDAC\PGCP-AI\assignment\Assignment -2> javap -c question26
Compiled from "question26.java"
class question26 {
  question26();
    Code:
       0: aload_0
       1: invokespecial #1                  // Method java/lang/Object."<init>":()V
       4: return

  public static void main(java.lang.String[]);
    Code:
       0: ldc           #2                  // String priyansu
       2: astore_1
       3: new           #3                  // class java/lang/Boolean
       6: dup
       7: aload_1
       8: invokespecial #4                  // Method java/lang/Boolean."<init>":(Ljava/lang/String;)V
      11: astore_2
      12: getstatic     #5                  // Field java/lang/System.out:Ljava/io/PrintStream;
      15: aload_2
      16: invokevirtual #6                  // Method java/io/PrintStream.print:(Ljava/lang/Object;)V
      19: return
}
```

Question 8b:

Code:

```
class question27 {

   public static void main(String[] args){

      byte b = 23;
      Byte c = Byte.valueOf(b);

      System.out.println(c);

   }

}
```

```
PS D:\CDAC\PGCP-AI\assignment\Assignment -2> javac question27.java
PS D:\CDAC\PGCP-AI\assignment\Assignment -2> java question27
23
```

Question 9:

Code:

```
class question28 {
 public static void main(String[] args) {

   Byte s1 = Byte.valueOf((byte) 120);

   byte b = s1.byteValue();
   short s = s1.byteValue();
   int i = s1.intValue();
   long l = s1.longValue();
   float f = s1.longValue();
   double d = s1.longValue();
   System.out.println("byte: " +b);
   System.out.println("short: " +s);
   System.out.println("int: " +i);
   System.out.println("long: " +l);
   System.out.println("float: " +f);
   System.out.println("double: " +d);
}
}
```

```
PS D:\CDAC\PGCP-AI\assignment\Assignment -2> javac question28.java
PS D:\CDAC\PGCP-AI\assignment\Assignment -2> java question28
byte: 120
short: 120
int: 120
long: 120
float: 120.0
double: 120.0
```

Question 8C:

Code:

```
class question29 {

   public static void main(String[] args){

      String A = "42";
      Byte B = Byte.valueOf(A);

      System.out.println("Byte Instance: " +B);

   }

}
```

```
PS D:\CDAC\PGCP-AI\assignment\Assignment -2> javac question29.java
PS D:\CDAC\PGCP-AI\assignment\Assignment -2> java question29
Byte Instance: 42
```

Question 13B;

Code:

```
class question30 {

   public static void main(String[] args) {

      short s1 = 34;
      Short s2 = Short.valueOf(s1);

      System.out.println(s2);

   }

}
```

```
PS D:\CDAC\PGCP-AI\assignment\Assignment -2> javac question30.java
PS D:\CDAC\PGCP-AI\assignment\Assignment -2> java question30
34
```

Question 13C;

Code:

```
class question31 {

   public static void main(String[] args) {

      String A1 = "342";

      short S1 = Short.parseShort(A1);

      System.out.println("Short Instance: " + S1);

   }

}
```

```
PS D:\CDAC\PGCP-AI\assignment\Assignment -2> javac question31.java
PS D:\CDAC\PGCP-AI\assignment\Assignment -2> java question31
Short Instance: 342
```

Question 14:

Code:

```
class question32 {

   public static void main(String[] args) {

      Short S1 = Short.valueOf((short)93);

      byte b = S1.byteValue();
      short s = S1.shortValue();
      int i = S1.intValue();
      long l = S1.longValue();
      float f = S1.floatValue();
      double d = S1.doubleValue();

      System.out.println("Byte: " +b);
      System.out.println("Short: " +s);
      System.out.println("Int: " +i);
      System.out.println("Long: " +l);
      System.out.println("Float: " +f);
      System.out.println("Double: " +d);

   }

}
```

```
PS D:\CDAC\PGCP-AI\assignment\Assignment -2> javac question32.java
PS D:\CDAC\PGCP-AI\assignment\Assignment -2> java question32
Byte: 93
Short: 93
Int: 93
Long: 93
Float: 93.0
Double: 93.0
```

Question 16b:

Code:

```
class question33 {

   public static void main(String[] args) {

int a = 45;

Integer b = a;

System.out.println(b);

   }

}
```

```
PS D:\CDAC\PGCP-AI\assignment\Assignment -2> javac question33.java
PS D:\CDAC\PGCP-AI\assignment\Assignment -2> java question33
45
```

Question 16c:

Code:

```
class question34 {

   public static void main(String[] args) {

      String ab = String.valueOf(83);

      Integer D = Integer.valueOf(ab);

      System.out.println(D);


   }

}
```

```
PS D:\CDAC\PGCP-AI\assignment\Assignment -2> javac question34.java
PS D:\CDAC\PGCP-AI\assignment\Assignment -2> java question34
83
```

Question 17:

Code:

```
class question35 {


   public static void main(String[] args) {

      Integer A = Integer.valueOf(83);


      byte b = A.byteValue();

      short s = A.shortValue();

      int i = A.intValue();

      long l = A.longValue();

      float f = A.floatValue();

      double d = A.doubleValue();


      System.out.println("Byte: " +b);

      System.out.println("Short: " +s);

      System.out.println("Integer: " +i);

      System.out.println("Long: " +l);

      System.out.println("Float: " +f);

      System.out.println("Double: " +d);


   }


}
```

```
PS D:\CDAC\PGCP-AI\assignment\Assignment -2> javac question35.java
PS D:\CDAC\PGCP-AI\assignment\Assignment -2> java question35
Byte: 83
Short: 83
Integer: 83
Long: 83
Float: 83.0
Double: 83.0
```

Question 20 B & C;

Code:

```
class question36 {


   public static void main(String[] args) {

      long X = 1495500L;

      Long L = Long.valueOf(X);


      System.out.println("Long value into Long Instance:" +L);


      String S1 = "342";

      Long L1 = Long.valueOf(S1);


      System.out.println("String Instance to Long Instance: " +L1);


   }


}
```

```
PS D:\CDAC\PGCP-AI\assignment\Assignment -2> javac question36.java
PS D:\CDAC\PGCP-AI\assignment\Assignment -2> java question36
Long value into Long Instance:1495500
String Instance to Long Instance: 342
```

Question 21:

Code:

```
class question37 {


   public static void main(String[] args){


      Long L1 = Long.valueOf(642392L);


      byte b = L1.byteValue();

      short s = L1.shortValue();

      int i = L1.intValue();

      long l = L1.longValue();

      float f = L1.floatValue();

      double d = L1.doubleValue();


      System.out.println("Byte:" +b);

      System.out.println("Short:" +s);

      System.out.println("Int:" +i);

      System.out.println("Long:" +l);

      System.out.println("Float:" +f);

      System.out.println("Double:" +d);

   }

}
```

```
PS D:\CDAC\PGCP-AI\assignment\Assignment -2> javac question37.java
PS D:\CDAC\PGCP-AI\assignment\Assignment -2> java question37
Byte:88
Short:-12968
Int:642392
Long:642392
Float:642392.0
Double:642392.0
```

Question 24 B & C;

Code:

```
class question38 {


   public static void main(String[] args) {


      float F1 = 43.43f;


      Float F2 = Float.valueOf(F1);


      System.out.println(F2);


      String S1 = String.valueOf("244");


      Float F3 = Float.valueOf(S1);


      System.out.println(F3);

   }

}
```

```
PS D:\CDAC\PGCP-AI\assignment\Assignment -2> javac question38.java
PS D:\CDAC\PGCP-AI\assignment\Assignment -2> java question38
43.43
244.0
```

Question 25:

Code:

```
class question39 {


   public static void main(String[] args) {


      Float F1 = Float.valueOf(75.85f);


      byte b = F1.byteValue();

      short s = F1.shortValue();

      int i = F1.intValue();

      long l = F1.longValue();

      float f = F1.floatValue();

      double d = F1.doubleValue();


      System.out.println("Byte:" +b);

      System.out.println("Short:" +s);

      System.out.println("Int:" +i);

      System.out.println("Long:" +l);

      System.out.println("Float:" +f);

      System.out.println("Double:" +d);

   }

}
```

```
PS D:\CDAC\PGCP-AI\assignment\Assignment -2> javac question39.java
PS D:\CDAC\PGCP-AI\assignment\Assignment -2> java question39
Byte:75
Short:75
Int:75
Long:75
Float:75.85
Double:75.8499984741211
```

Question 28 B & C;

Code:

```
class question40 {


   public static void main(String[] args) {


      double D1 = 123;

       double D2 = Double.valueOf(D1);


       System.out.println(D2);


      String S1 = "232";

      double D3 =  Double.valueOf(S1);


      System.out.println(D3);


   }


}
```

```
PS D:\CDAC\PGCP-AI\assignment\Assignment -2> javac question40.java
PS D:\CDAC\PGCP-AI\assignment\Assignment -2> java question40
123.0
232.0
```

Question 29:

Code:

```
class question41 {


   public static void main(String[] args) {
      Double D1 = Double.valueOf(324.3);


      byte b = D1.byteValue();
      short s = D1.shortValue();
      int i = D1.intValue();
      long l = D1.longValue();
      float f = D1.floatValue();
      double d = D1.doubleValue();


      System.out.println("Byte:" +b);
      System.out.println("Short:" +s);
      System.out.println("Int:" +i);
      System.out.println("Long:" +l);
      System.out.println("Float:" +f);
      System.out.println("Double:" +d);
   }


}
```

```
PS D:\CDAC\PGCP-AI\assignment\Assignment -2> javac question41.java
PS D:\CDAC\PGCP-AI\assignment\Assignment -2> java question41
Byte:68
Short:324
Int:324
Long:324
Float:324.3
Double:324.3
```